

C01-AT1

Short Answer Test

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Course
Name : Computer Vision

Course
code : ITA0501

Slot : A

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1) Here, the main problem is very low spatial resolution which leads small objects to appear unclear. To overcome this

"Sampling" concept is applied which determines the number of pixels used to represent an image and makes increase of number of pixels.

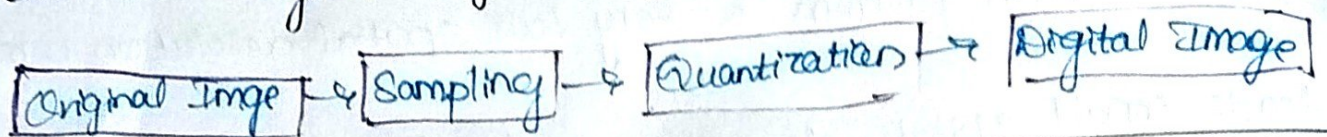
Increasing the sampling rate or using high-resolution camera will capture more details and improve image clarity even for the small objects.

2) For a dark or noisy night image, use of histogram equalization image-processing technique is the best as it improve the brightness and contrast. Gaussian filter can be used to reduce the noise and make image more clear. These techniques make objects more visible and improve the quality of the image.

3) Continuous intensity values can be converted into digital values using Quantization.

"Quantization" divides the continuous intensity range into a fixed number of discrete levels. This allows the image to be stored, processed, analysed and displayed

as the image. (a) digital image.



4) Uneven lighting can make a image of the object too dark or too bright based on the lighting. Here Histogram Equalization or contrast enhancement can be applied to improve the product and defect visibility. It improves the intensity differences.

This makes defects such as scratches or cracks more clearly visible.

5) Here, "Image Segmentation" can be used to divide the image into regions such as roads, vehicles, pedestrians, obstacles and the background. It helps the car to identify the objects and understand its surroundings.

This information supports safe navigation and driving decisions.

6) Edge Detection, such as Canny edge detector, can be used to identify cracks in manufactured components. Cracks usually create sudden changes in intensity and appear as edges in the image.

Detecting these edges helps the system identify defective products automatically.

7) A Median filter can be used to remove unwanted noise from the medical image. It replaces noisy pixels using the median value of their neighboring pixels. This reduces noise while preserving important edges, making the tumor easier to identify.

8) Object detection can identify vehicles in traffic images, while object tracking follows their movement across video frames. This helps in counting vehicles and monitoring their movement. It is useful for detecting congestion and improving traffic management.

9) The blur in the image is mainly caused by the movement of the object or camera during the image acquisition, especially when the shutter speed is slow. Here a faster shutter speed can reduce motion blur during image capture.

Image deblurring or sharpening techniques can be applied

to improve the quality of the captured image.

10. Image classification can be used to classify manufactured products as good or defective. A model is trained using both the images good and defective product and learns their visual features. It can then automatically classify new products, helping in faster and more reliable quality inspection.

